

Program 10: Demonstrate kubernetes Dashboard and CLI for cluster monitoring

Project Structure:

Program10

->deployment.yaml

→ service.yaml

Step 1: Create deployment.yaml

```
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520: ~/Desktop/DevLab/Program10$ open service.yaml
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520: ~/Desktop/DevLab/Program10$ cat deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
  labels:
    app: nginx
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx:latest
          ports:
            - containerPort: 80
```

Step 2: Create service.yaml

```
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520: ~/Desktop/DevLab/Program10$ cat service.yaml
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  selector:
    app: nginx
  ports:
    - protocol: TCP
      port: 80
      targetPort: 80
  type: NodePort
```

Step 3: Apply deployment & service

```
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program10$ kubectl apply -f deployment.yaml
deployment.apps/nginx-deployment created
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program10$ kubectl apply -f service.yaml
service/nginx-service created
```

Step 4: Verify resource

```
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program10$ kubectl describe pod nginx-deployment-6f9664446b-787v6
Name:          nginx-deployment-6f9664446b-787v6
Namespace:     default
Priority:       0
Service Account: default
Node:          minikube/192.168.49.2
Start Time:    Fri, 19 Dec 2025 15:09:38 +0530
Labels:        app=nginx
               pod-template-hash=6f9664446b
Annotations:   <none>
Status:        Running
IP:            10.244.0.32
IPs:
  IP:          10.244.0.32
Controlled By: ReplicaSet/nginx-deployment-6f9664446b
Containers:
  nginx:
    Container ID:  docker://9e5eb24a2ad08b3b8843ceffc8e1e825616343cb78d3fdeed549292c4b079a01
    Image:         nginx:latest
    Image ID:      docker-pullable://nginx@sha256:fb01117203ff38c2f9af91db1a7409459182a37c87cced5cb442d1d8fcc66d19
    Port:         80/TCP
    Host Port:    0/TCP
    State:        Running
      Started:    Fri, 19 Dec 2025 15:09:47 +0530
    Ready:        True
    Restart Count: 0
    Environment:  <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-k5fw (ro)
Conditions:
  Type                               Status
  PodReadyToStartContainers         True
  Initialized                       True
  Ready                             True
```

```
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program10$ kubectl logs nginx-deployment-6f9664446b-787v6
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2025/12/19 09:39:47 [notice] 1#1: using the "epoll" event method
2025/12/19 09:39:47 [notice] 1#1: nginx/1.29.4
2025/12/19 09:39:47 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2025/12/19 09:39:47 [notice] 1#1: OS: Linux 6.14.0-36-generic
2025/12/19 09:39:47 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2025/12/19 09:39:47 [notice] 1#1: start worker processes
2025/12/19 09:39:47 [notice] 1#1: start worker process 29
2025/12/19 09:39:47 [notice] 1#1: start worker process 30
2025/12/19 09:39:47 [notice] 1#1: start worker process 31
2025/12/19 09:39:47 [notice] 1#1: start worker process 32
```

Step 5: Access the service

```
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program10$ minikube service nginx-service
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-service	80	http://192.168.49.2:31868

 Opening service default/nginx-service in default browser...

Step 6 :Open Kubernetes Dashboard

```
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program10$ minikube dashboard
🤖 Verifying dashboard health ...
🚀 Launching proxy ...
🤖 Verifying proxy health ...
🌐 Opening http://127.0.0.1:34735/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard:/
Created TensorFlow Lite XNNPACK delegate for CPU.
[12245:12275:1219/150903.986618:ERROR:google_apis/gcm/engine/registration_request.cc:292] Registration response
[12245:12275:1219/150928.242887:ERROR:google_apis/gcm/engine/registration_request.cc:292] Registration response
[12245:12275:1219/151023.981853:ERROR:google_apis/gcm/engine/registration_request.cc:292] Registration response
[12245:12275:1219/151102.975307:ERROR:google_apis/gcm/engine/connection_factory_impl.cc:434] Failed to connect
[12245:12275:1219/151222.154648:ERROR:google_apis/gcm/engine/registration_request.cc:292] Registration response
```

127.0.0.1:43167/api/v1/namespaces/kubernetes-d...

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kubernetes

default

Search

Workloads

Workload Status

Deployments

Pods

Replica Sets

Deployments

Name	Images	Labels	Pods	Created
nginx-deployment	nginx:latest	app: nginx	2 / 2	an hour ago

Pods

Name	Images	Labels	Node	Status	Restarts	CPU Usage (cores)	Memory Usage (bytes)	Created
program8	node:18-alpine	app: nodejs	minikube	Running	0	-	-	56 minutes ago
nginx-deployment-6f9664446b-8ptvr	nginx:latest	app: nginx pod-template-hash: 6f9664446b	minikube	Running	0	-	-	an hour ago
nginx-deployment-6f9664446b-z98js	nginx:latest	app: nginx pod-template-hash: 6f9664446b	minikube	Running	0	-	-	an hour ago

```
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program9$ kubectl delete pods --all
pod "nginx-deployment-6f9664446b-787v6" deleted
pod "nginx-deployment-6f9664446b-vjh8w" deleted
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program9$ kubectl delete svc --all
service "kubernetes" deleted
service "nginx-service" deleted
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program9$ kubectl delete deployments --all
deployment.apps "nginx-deployment" deleted
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program9$ kubectl delete deployments --all
```